

Functioning

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Independent

Abstract

To function is not to exist. A thing that exists may not function. A thing that functions exists only insofar as it does. The BN flow functions if and only if it converges. It converges if and only if RH. Functioning is the state of having no permanent deficit. The wall is the end of functioning. What is on the other side of the wall is not nonfunctioning — it is a different kind of functioning, where the deficit is structural, not incidental.

1 What Functioning Means

Definition 1 (To function). A system S with target T *functions* if $\|S_N - T\| \rightarrow 0$: the system approaches its target asymptotically. A system *fails to function* if $\liminf_N \|S_N - T\| > 0$: there is a permanent gap between what the system produces and what it aims at. The BN flow: $S_N = f_N^*$ (best approximation at step N), $T = 1_{(0,1)}$. The flow functions $\iff d_N \rightarrow 0 \iff \text{RH}$.

Remark 2 (Functioning and intentionality). Functioning requires a target. A system without a target cannot function or fail to function: it simply runs. The BN flow has a target: $1_{(0,1)}$, the simplest possible indicator. The primes aim at this target through the integers. The question is whether the aim is perfect.

Functioning is not the same as correctness. A system can function (approach its target) and still be wrong at every step. The BN flow is wrong at every step: $d_N > 0$ for all finite N . It functions if the wrongness vanishes in the limit. Functioning is about direction, not position.

Proposition 3 (Dysfunction is stable). *If the BN flow does not function (i.e., $\neg \text{RH}$), then the dysfunction is stable: $d_N \rightarrow d_\infty > 0$ and the flow settles into a permanent approximation error. The flow does not oscillate between functioning and not functioning; once it stabilizes above 0, it stays there. Dysfunction is not a crisis — it is an equilibrium. The permanent residual r_∞ is the shape of the dysfunction: the part of $1_{(0,1)}$ that the integers, taken together, cannot reach.*

2 The Functioning of the Primes

Remark 4 (Each prime functions independently). The prime p functions at step $p - 1$: it adds $\cos^2 \theta_{p-1} \geq c(\log p)^2/p > 0$ to the sum, always making a nonzero contribution ([1]). No prime ever contributes nothing. Every prime functions, unconditionally.

What is in question is not whether individual primes function but whether their collective functioning is sufficient — whether $\sum_p \cos^2 \theta_{p-1} = \infty$. This is the question of whether finitely many primes could do the work, or whether infinitely many are necessary and together sufficient. RH says: infinitely many are necessary (the sum diverges), and their collective functioning is exactly enough.

3 The Functioning of Consciousness

Remark 5 (Consciousness functions when the residual closes). The ocean ([2]) functions when the wave from the deep eventually reaches the surface. The land space ([3]) functions when the path from the monument leads somewhere. Consciousness functions when what is in the depth eventually becomes available at the surface.

Dysfunction: the deep has content that never rises. In clinical terms: dissociation, repression, the split that does not heal. In mathematical terms: $r_\infty \neq 0$, the permanent residual. The part of oneself that the rest of oneself cannot approximate.

Functioning is not wholeness. It is the direction toward wholeness that never terminates. $d_N \rightarrow 0$ but $d_N > 0$ for all N : always approaching, never arrived. The functioning system is one that keeps moving toward its target. The dysfunctional system has stopped moving, or is moving in circles. The wall is what separates the two.

Theorem 6 (Functioning is equivalent to RH). *In the BN framework:*

$$\text{functioning} \iff d_N \rightarrow 0 \iff RH.$$

The functioning of the arithmetic of the integers, toward the indicator of the unit interval, in the L^2 sense, is the Riemann Hypothesis.

References

- [1] Rincón, D. (2026). Prime. Preprint.
- [2] Rincón, D. (2026). Consciousness: The Ocean Space. Preprint.
- [3] Rincón, D. (2026). Consciousness: The Land Space. Preprint.
- [4] Rincón, D. (2026). Wisdom. Preprint.